PATHOLOGY

Diseases of Vulva, Vagina & Cervix

LG 2
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GUE BLOCK

Pathology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What distinguishes an intraepithelial lesion (CIN/SIL) from invasive carcinoma on histology?
2. Which two HPV oncoproteins drive cervical carcinogenesis, and which tumour suppressors do they target?
3. Name the organism pair most responsible for infectious vaginitis, and the pair most responsible for infectious cervicitis.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Vulva

Vulvitis — inflammatory & infectious

Type	Cause / features
Noninfectious	Irritant or allergic contact dermatitis; erythema, pruritus, discomfort.
Infectious	*Candida* species, HSV, HPV-related lesions, sexually transmitted bacterial infections.

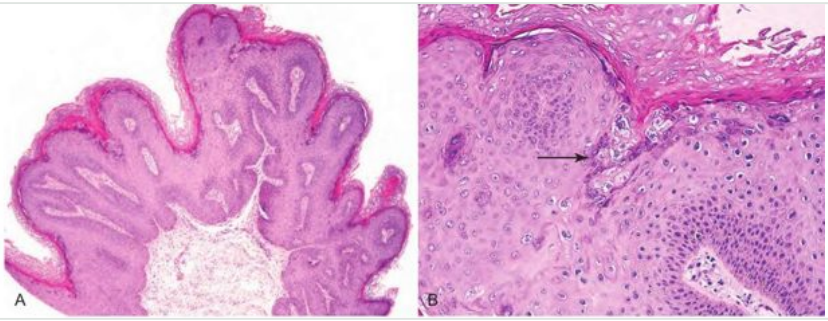
Histology varies by cause — superficial inflammation up to squamous intraepithelial lesions. **Persistent vulvar pruritus or abnormal white plaques require evaluation** for chronic dermatoses or neoplasia.

Bartholin duct cyst & abscess

Obstruction of a Bartholin gland duct → retention of secretions → **duct cyst** (often painless). Secondary infection → painful **abscess**.

Condyloma acuminatum (genital wart) — illness script

Nature	**Benign** exophytic squamous proliferation, most often **low-risk HPV 6 and 11**. Transmitted by sexual contact.
Gross	Soft, papillary or verrucous lesion.
Microscopy	Fibrovascular papillae, acanthosis, hyperkeratosis, **koilocytosis**. Koilocytes show perinuclear clearing and enlarged, irregular, hyperchromatic nuclei.



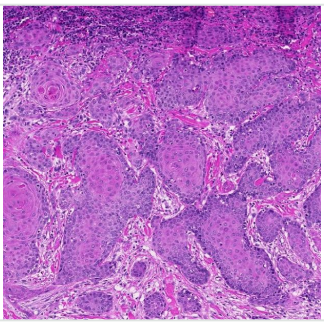
Condyloma acuminatum — exophytic papillary architecture (A); koilocytic atypia with perinuclear halos (B, arrow)

Carcinoma of vulva

90% are squamous cell carcinomas; the rest are mainly adenocarcinomas or basal cell carcinomas.

Pathway	Features
HPV-associated	Usually high-risk HPV (particularly **HPV 16**); arises through **HPV-associated vulvar intraepithelial neoplasia (VIN)**.
HPV-independent	Often associated with **lichen sclerosus** and HPV-independent VIN.

Gross: ulcerated, nodular, or exophytic lesion. **Histology:** invasive squamous nests.

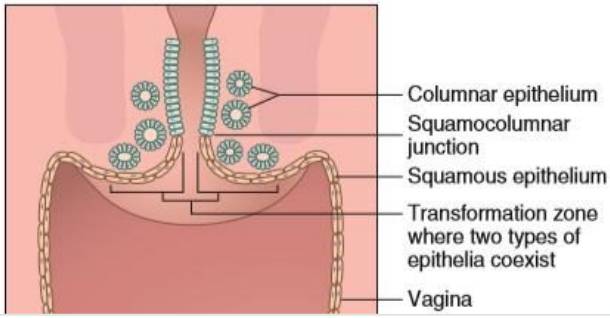


Vulval invasive squamous cell carcinoma — invasive nests with keratin pearls

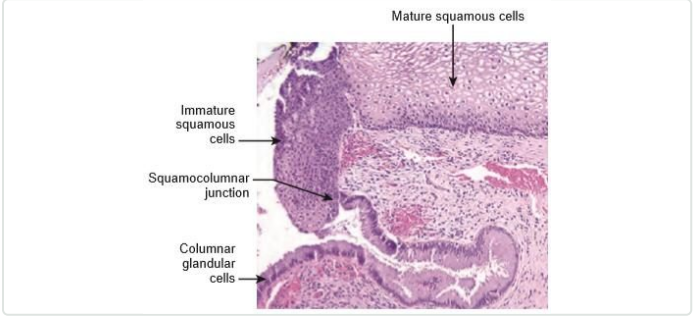
2 Vagina

Vaginitis	Relatively common, usually transient, **of no clinical consequence**. Many organisms implicated; frequent offenders are ***Candida albicans*** and ***Trichomonas vaginalis***.
Vaginal carcinoma	**Primary vaginal carcinoma is rare** — usually extension or metastasis from cervix or vulva. Most primary vaginal cancers are **squamous cell carcinoma**, often HPV-associated. **Vaginal HSIL (VaIN)** is an important precursor. Typically presents in older adults with vaginal bleeding or a visible lesion.

3 Cervix



Squamocolumnar junction & transformation zone



Transformation zone histology — mature/immature squamous cells, squamocolumnar junction, columnar glandular cells

Cervicitis

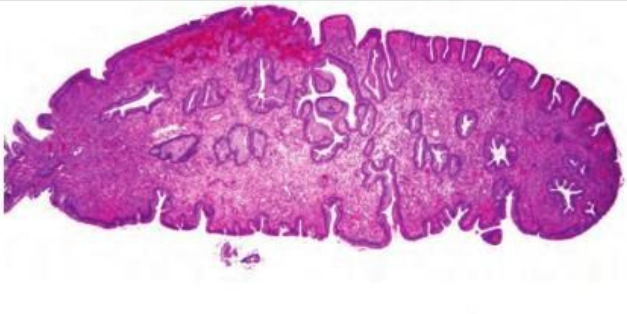
Subclassified as **infectious (specific) or non-infectious (non-specific)**, and as **acute or chronic**.

Type	Finding
Acute cervicitis	**Neutrophils** in endocervical epithelium and stroma.
Chronic cervicitis	**Lymphocytes and plasma cells**; often nonspecific.

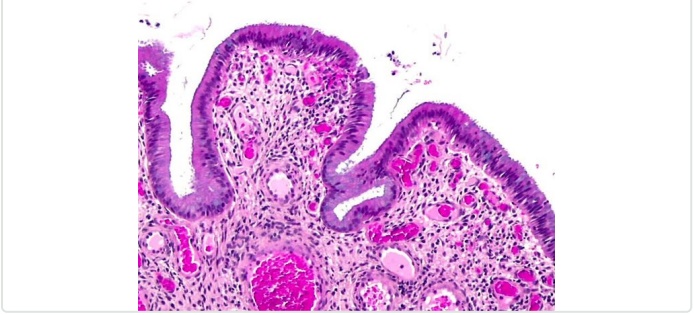
Important infectious causes: ***Chlamydia trachomatis*** and ***Neisseria gonorrhoeae***; HSV and *Trichomonas* may also involve the cervix.

Endocervical polyp

Benign polypoid proliferation of endocervical mucosa, protruding from the endocervical mucosa (sometimes from the external os). **Microscopy:** mucinous endocervical glands in edematous or fibrous, vascular stroma, often inflamed. May cause intermenstrual or postcoital spotting; many asymptomatic.



Endocervical polyp — gross, dense fibrous stroma



Endocervical columnar epithelium with mucous secretion

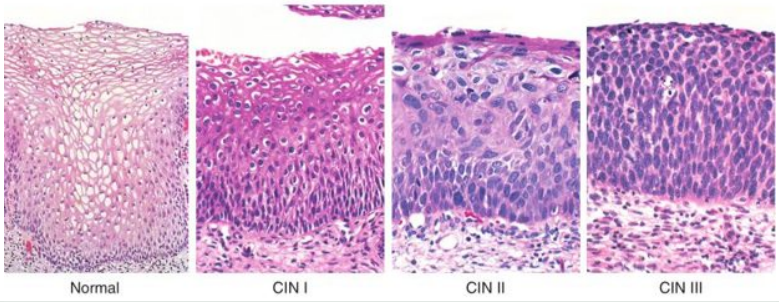
Cervical squamous intraepithelial lesions

Persistent infection by oncogenic HPV, particularly **types 16 and 18**, underlies most cervical carcinomas. HPV enters immature squamous cells, especially in the transformation zone. An intraepithelial lesion has an **intact basement membrane** — invasion defines carcinoma.

Dysplasia / CIS	CIN	SIL
Mild dysplasia	CIN I	**LSIL**
Moderate dysplasia	CIN II	**HSIL**
Severe dysplasia	CIN III	
Carcinoma in situ	CIN III	

Histologic grading — the discriminator

Grade	Extent of atypia
CIN 1 / LSIL	Atypia mainly in **basal one-third**; maturation preserved; koilocytosis may occur.
CIN 2 / HSIL	Atypical immature cells extend into **middle epithelial layers**; mitoses may occur above basal layer.
CIN 3 / HSIL	**Severe atypia and loss of maturation** through most or all epithelial thickness.



CIN spectrum: Normal → CIN I (basal atypia) → CIN II (two-thirds) → CIN III (full-thickness, carcinoma in situ)

Cervical cancer screening & prevention

- **Cervical cytology (Pap testing)** detects abnormal exfoliated squamous cells.
- **High-risk HPV testing** detects oncogenic viral types.
- Abnormal screening results may require **colposcopy with biopsy** for histologic diagnosis.
- Screening recommendations vary with age, risk, and country.
- Population-based **HPV vaccination and screening** reduce cervical cancer burden.

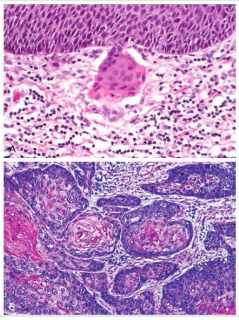
Cervical carcinoma

Histologic type	Frequency
Squamous cell carcinoma	**~80%** — most common subtype
Adenocarcinoma	**~15%**
Adenosquamous & neuroendocrine	Rare — remaining ~5%

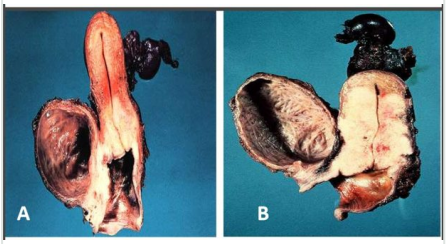
Pathogenesis	Persistent high-risk HPV infection is the major driver of cervical squamous carcinoma and many adenocarcinomas. Viral **E6** promotes degradation of **p53**; **E7** functionally inactivates **RB**.
Epidemiology	Peak incidence of invasive cervical carcinoma is **45 years**.
Clinical behavior	Early lesions may be asymptomatic; **abnormal vaginal bleeding, especially postcoital**, is a common presentation. Local extension may involve vagina, parametria, bladder, or rectum; lymphatic spread involves pelvic nodes.
Prognosis	**Stage at diagnosis is the strongest prognostic factor**; nodal disease and histologic type also matter. Small-cell neuroendocrine carcinoma is uncommon and typically clinically aggressive. Most stage IV deaths result from **local extension** rather than distant metastases.



Invasive, exophytic cervical carcinoma at the os



A: microinvasive SCC breaking through basement membrane of HSIL. B: invasive SCC



Cervical SCC — A: extension into bladder. B: massive extension into uterine corpus

P Must-remember pearls

- An intraepithelial lesion has an **intact basement membrane** — invasion is what defines carcinoma.
- CIN grade tracks **how far up the epithelium** atypia and loss of maturation extend: basal third (CIN1) → two-thirds (CIN2) → full thickness (CIN3/CIS).
- HPV **E6 degrades p53**; HPV **E7 inactivates RB** — the two-hit mechanism behind cervical carcinogenesis.
- Vulvar SCC has **two pathways**: HPV-associated (HPV 16, via VIN) and HPV-independent (lichen sclerosus).
- Primary vaginal carcinoma is **rare** — most vaginal cancers are secondary extension from cervix or vulva.
- Cervical SCC (~80%) > adenocarcinoma (~15%) > adenosquamous/neuroendocrine (~5%).
- Stage IV cervical cancer mortality is driven mainly by **local extension**, not distant metastasis.

Q Test yourself

1. A 28-year-old presents with soft, papillary vulvar lesions. Biopsy shows fibrovascular papillae with koilocytic atypia. Which HPV types are most associated with this lesion?

A. HPV 16 and 18
B. HPV 6 and 11
C. HPV 31 and 33
D. HPV 45 and 52

2. A cervical biopsy shows atypical immature squamous cells confined to the basal one-third of the epithelium, with preserved maturation above. This corresponds to:

A. CIN 1 / LSIL
B. CIN 2 / HSIL
C. CIN 3 / HSIL
D. Invasive squamous cell carcinoma

3. Which histologic finding is required to distinguish invasive cervical carcinoma from a high-grade squamous intraepithelial lesion?

A. Degree of nuclear atypia
B. Presence of koilocytes
C. Breach of the basement membrane
D. Mitotic rate

4. A 60-year-old woman presents with a vaginal mass. Which statement about her diagnosis is most accurate?

A. Primary vaginal carcinoma is the most common cause of a vaginal mass at this age
B. A vaginal mass in this age group is more likely to represent extension from cervix or vulva than a primary vaginal tumor
C. Vaginal carcinomas are rarely associated with HPV
D. Vaginal HSIL (VaIN) has no relationship to vaginal carcinoma

5. Regarding the molecular pathogenesis of cervical squamous carcinoma, which pairing is correct?

A. E6 inactivates RB; E7 degrades p53
B. E6 degrades p53; E7 inactivates RB
C. E6 and E7 both target p53 exclusively
D. E6 and E7 act only on BRCA1

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 · B — Condyloma acuminatum (genital wart) is most often caused by low-risk HPV 6 and 11.

2 · A — Atypia confined to the basal third with preserved maturation defines CIN 1 / LSIL.

3 · C — An intraepithelial lesion has an intact basement membrane; invasion through it defines carcinoma.

4 · B — Primary vaginal carcinoma is rare — most vaginal tumors represent extension or metastasis from the cervix or vulva.

5 · B — HPV E6 promotes degradation of p53; E7 functionally inactivates RB.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam